

Abhinandan

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EXPERIENCE

Founding Software Engineer

Sept 2025 – Present

Browser

Remote

- Built Browser's Chrome MV3 recorder + CDP-native browser automation agent, achieving 95%+ precise AX/DOM element capture with cross-iframe support, obstruction checks, and real mouse/key/upload execution.
- Built a smart streaming ReAct loop across FastAPI + extension with SSE tool execution, multi-tab orchestration, safe parallelism, abort/continue, and audit logs.
- Cut automation LLM spend by roughly 67% using compact recording traces, context-window compression, prompt caching, and model-routing across GPT-5, Claude Sonnet & Haiku.
- Shipped a zero-LLM replay engine: recordings run as variable-driven tool-call templates, with a stateful AI fallback that resumes mid-run on failure.
- Shipped self-healing docs that auto-repair on UI drift — Haiku → Sonnet diff triage, LLM-free replay of intact steps, and a CDP agent that fixes only what changed.

Founding Software Engineer

Jan 2025 – Jul 2025

Cynos Nexus

Noida, India

- Shipped core features of an AI-powered real estate platform using Next.js, Nest.js, GraphQL, Redis, and GCP.
- Built the AI knowledge base service using FastAPI, LangChain, and vector retrieval pipelines, powering customer-facing search workflows.
- Developed document-ingestion pipelines using Google Cloud Vision, XLSX processing, and BullMQ workers, enabling automated extraction of customer data from spreadsheets and scanned records.
- Automated containerized CI/CD infrastructure via Docker, GitHub Actions, and Nginx for reverse proxy/load balancing.

Contract Software Engineer

Sept 2024 – Dec 2024

Etkin.ai

Remote, Turkiye

- Built a LangChain + pgvector knowledge base powering AI-assisted document search and retrieval workflows, improving query accuracy by 15%.
- Developed scalable data-ingestion pipelines using bulk CSV processing and Celery workers, reducing processing time by 40%.
- Engineered a production PDF generation system transforming structured AI outputs and dynamic JSON reports into enterprise-grade documents.
- Automated deployment of AI services using Docker, GitHub Actions, and AWS EC2, establishing reliable CI/CD workflows for production environments.

Open Source Contributor

June 2024 – Aug 2024

HeroUI

Y Combinator S24 startup

- Received a **personal offer from the CEO** to join HeroUI (prev. NextUI) after making **open-source contributions**.
- Resolved 10+ bugs & delivered 7+ feature enhancements in core components including Calendar, Table and Pagination.

SWE Intern

Jan 2024 – Apr 2024

SkilledUp

Noida

OPEN SOURCE PROJECTS

AgentFlow-Pro (*Research*) | *PyTorch, TRL, DAPO, PRM, PEFT/LoRA, Ollama, FastMCP* | github.com/awesome-pro/agentflow-pro

- Rebuilt and extended the ICLR 2026 *AgentFlow* architecture as a local Qwen3-8B Planner → Executor → Verifier → Memory loop (grammar-constrained planning, Tavily search, sandboxed Python/SymPy tools); trained the planner with **DAPO + a learned Qwen3-0.6B Process Reward Model** (replacing the paper's GRPO) and a hand-built dynamic-sampling stage TRL lacks, then ran the full pipeline on an A40 (DeepSeek-judge step labels → PRM → bf16 LoRA DAPO → GGUF) and measured **+5.0 pts on GPQA-Diamond (40.0%→45.0%)** under leakage-free, quantization-matched evaluation.

guardloop | *OpenAI/Anthropic SDKs, OpenTelemetry, LangGraph* | github.com/awesome-pro/guardloop

- Production guardrail runtime for async agents: pre-flight Decimal cost/token/time/tool-call budgets, per-tool circuit breakers, verifier feedback retry loop, OpenTelemetry GenAI spans, and no-rewrite LangGraph/OpenAI Agents SDK adapters that enforce safety inside existing LLM/tool calls.

smartmemo | *PyTorch, sentence-transformers, FAISS, SQLite* | github.com/awesome-pro/smartmemo

- Semantic memory/cache for LLM agents where embeddings retrieve candidates and a learned pair-equivalence classifier decides reuse; shipped bundled classifier-v2 trained on 16,576 pairs across 9 domains (+30 precision points at equal recall vs cosine), plus feedback export, implicit bad-hit detection, gated retraining, and WAL-backed SQLite hardening.

orchflow | *Asyncio, LiteLLM* | github.com/awesome-pro/orchflow

- Dependency-free typed Python framework for readable multi-agent pipelines: sequential/parallel/conditional flows, retries, shared **StepContext**, flat traces, lifecycle events, human gates, JSON checkpoint/resume, and optional LiteLLM Agent with structured JSON/Pydantic outputs.

agenteval | *AsyncIO, OpenAI/Anthropic/LangChain adapters* | github.com/awesome-pro/agenteval

- Agent evaluation toolkit replacing exact-match asserts with repeated-run pass-rate tests; traces tool calls/timing/steps, supports collect-then-raise behavioral assertions, OpenAI/Anthropic/LangChain adapters, and Typer CLI JSON reports for CI gates.

TECHNICAL SKILLS

ML & Retrieval: PyTorch, Hugging Face, sentence-transformers, embeddings, classifier training, RAG, FAISS, Pinecone, Qdrant, Weaviate, ChromaDB, pgvector

AI Agents & Frameworks: LangGraph, OpenAI Agents SDK, LangChain, LlamaIndex, CrewAI, AutoGen, Pydantic AI, LiteLLM, MCP / FastMCP, multi-agent orchestration, ReAct, function calling, tool use, CDP-based browser agents

LLM Engineering: Fine-tuning (LoRA, QLoRA, PEFT), RLHF / DPO / GRPO / DAPO / PRM, quantization (AWQ, GPTQ, GGUF), inference engines (vLLM, SGLang, Ollama, TGI), context engineering, KV-cache strategies, streaming

Observability & Eval: OpenTelemetry, Langfuse, Arize Phoenix, LangSmith, statistical pass-rate evaluation, AIME / GPQA benchmarks

Cloud & Infra: Docker, GCP (Vertex AI), AWS, Nginx, Redis, PostgreSQL, MongoDB

Backend: FastAPI, Nest.js, Node.js, GraphQL, WebSockets, REST, SSE

Languages: Python, TypeScript, C++, SQL, Bash

Frontend: Next.js, React, Redux, Turbopack, Edge Runtime

AWARDS & RECOGNITION

International Youth Math Challenge Gold Honour | **Top 1% TypeScript Engineer Globally** (Algora) | **Amazon ML Summer School 2025** | 2nd Place, Outlier AI Hackathon

Reliance Foundation Scholar | **HDFC Badhate Kadam Scholar**

EDUCATION

Dr. APJ Abdul Kalam Technical University

B.Tech, Computer Science | Nov 2022 – Apr 2026